#1

Contains Duplicate

Arrays & Hashing | LeetCode 217 | NeetCode "Duplicate Integer"

PROBLEM

Given an integer array `nums`, return `true` if any value appears at least twice in the array, and return `false` if every element is distinct.

```
nums = [1,2,3,1] → true
nums = [1,2,3,4] → false
nums = [1,1,1,3,3,4,3,2,4,2] → true
```

CORE IDEA

Goal

Determine whether any value repeats in the array.

Key insight

A hash set gives O(1) average-case lookup and insertion, so a single pass can catch a repeat the moment it's seen, with no nested loops.

> Trigger: “Have I seen this value before?” → reach for a Hash Set membership check.

OPTIMAL SOLUTION

`Data structure` — Hash Set (Python `set`)

1. Initialize an empty set `h`.
2. Iterate through each `num` in `nums`.
3. If `num` is already in `h`, return `True` immediately — duplicate found.
4. Otherwise, add `num` to `h` and continue.
5. If the loop finishes with no match, return `False`.

O(n)	O(n)
TIME	SPACE

CODE

```python
class Solution:
    def hasDuplicate(self, nums: List[int]) ->
bool:
        h = set()
        for num in nums:
            if num in h:
                return True
            else:
                h.add(num)

        return False
```

TIPS TO REMEMBER

`Mental shortcut` — “Seen it before? Set it and forget it.”

`Common / related` — Contains Duplicate II (nearby-index variant), Contains Duplicate III (nearby-value variant), Longest Consecutive Sequence, Single Number

> *"I'm trading O(n) space for O(n) time with a hash set, which avoids the O(n²) brute-force approach of comparing every pair."*

EDGE CASES

- Empty array → no duplicates possible, return False
- All elements identical → returns True as soon as the second element is checked
- Single-element array → return False
- Negative numbers and zero → ints are hashable as-is, no special handling needed